SPARKCURV HR PORTAL

Installation Guide for AWS EC2 (Ubuntu)

Step-by-step deployment guide for production server

Layer	Technology
Frontend	React 18 + Tailwind CSS + Shadcn UI
Backend	Python 3.11 + FastAPI + Gunicorn
Database	MySQL 8.0 / MariaDB 10.5+
Web Server	Nginx (reverse proxy)
Auth	JWT (httpOnly cookies) + bcrypt
PDF/Excel	ReportLab + OpenPyXL + num2words

Step 1: Launch EC2 Instance

1. Go to AWS Console → EC2 → Launch Instance
2. Choose **Ubuntu 22.04 LTS** (or 24.04 LTS)
3. Instance type: **t2.micro** (free tier) or **t2.small** for production
4. Security Group — open these ports:

Port	Protocol	Source	Purpose
22	TCP	Your IP	SSH access
80	TCP	0.0.0.0/0	HTTP
443	TCP	0.0.0.0/0	HTTPS (SSL)

5. Download your .pem key file and connect:

```
ssh -i your-key.pem ubuntu@
```

Step 2: Install System Dependencies

```
sudo apt update && sudo apt upgrade -y

# Install Node.js 18+
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs

# Install Python 3.11+, pip, venv
sudo apt install -y python3 python3-pip python3-venv

# Install MariaDB
sudo apt install -y mariadb-server mariadb-client

# Install Nginx
sudo apt install -y nginx

# Install Yarn
sudo npm install -g yarn

# Install Git
sudo apt install -y git
```

Verify installations:

```
node -v # Should be 18+
python3 --version # Should be 3.11+
mysql --version # Should be 10.5+
nginx -v
```

Step 3: Setup MariaDB Database

```
# Start MariaDB
sudo systemctl start mariadb
sudo systemctl enable mariadb

# Secure installation (set root password)
sudo mysql_secure_installation

# Login and create database
sudo mysql -u root -p
```

Run these SQL commands:

```
CREATE DATABASE hr_portal;
CREATE USER 'hruser'@'localhost' IDENTIFIED BY 'your_strong_password';
GRANT ALL PRIVILEGES ON hr_portal.* TO 'hruser'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

■ *Replace 'your_strong_password' with a secure password. You can also use root user directly.*

Step 4: Clone the Repository

```
cd /var/www
sudo mkdir sparkcurv && sudo chown $USER:$USER sparkcurv
cd sparkcurv
git clone .

# Or if repo has separate folders:
# git clone hr-portal
# cd hr-portal
```

Step 5: Setup Backend

```
cd /var/www/sparkcurv/backend

# Create virtual environment
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt
pip install num2words gunicorn
```

Create backend/.env file:

```
nano .env
```

Paste this content (update values):

```
MYSQL_HOST=127.0.0.1
MYSQL_PORT=3306
MYSQL_USER=root
MYSQL_PASSWORD=your_db_password
MYSQL_DB=hr_portal

JWT_SECRET=generate-a-64-char-random-string-here
ADMIN_EMAIL=ponish.jino@sparkcurv.com
ADMIN_PASSWORD=Aiden@1996
FRONTEND_URL=https://hr.sparkcurv.in
```

■ *Generate JWT_SECRET with: python3 -c "import secrets; print(secrets.token_hex(32))"*

■ *Set FRONTEND_URL to your actual domain (for cookie security on HTTPS)*

Test backend runs:

```
source venv/bin/activate
python3 -c "from server import app; print('Backend OK')"
# If no errors, backend is ready
```

Step 6: Setup Frontend

```
cd /var/www/sparkcurv/frontend

# Install dependencies
yarn install

# Create frontend/.env file
nano .env
```

Paste this content:

```
REACT_APP_BACKEND_URL=https://hr.sparkcurv.in
```

■ *Replace with your actual domain or EC2 IP address (http://your-ip)*

Build for production:

```
yarn build

# This creates the /build folder with static files
```

Step 7: Create Systemd Service (Backend)

```
sudo nano /etc/systemd/system/hr-backend.service
```

Paste this content:

```
[Unit]
Description=HR Backend FastAPI
After=network.target

[Service]
User=www-data
Group=www-data
WorkingDirectory=/var/www/sparkcurv/backend
Environment="PATH=/var/www/sparkcurv/backend/venv/bin"
ExecStart=/var/www/sparkcurv/backend/venv/bin/gunicorn \
-k uvicorn.workers.UvicornWorker \
-w 4 --bind 127.0.0.1:8005 server:app
Restart=always

[Install]
WantedBy=multi-user.target
```

```
# Fix permissions
sudo chown -R www-data:www-data /var/www/sparkcurv/

# Start the service
sudo systemctl daemon-reload
sudo systemctl enable hr-backend
sudo systemctl start hr-backend

# Check status
sudo systemctl status hr-backend
```

Step 8: Configure Nginx

```
sudo nano /etc/nginx/sites-available/hr-portal
```

Paste this configuration:

```
server {
listen 80;
server_name hr.sparkcurv.in; # Your domain or EC2 IP

root /var/www/sparkcurv/frontend/build;
index index.html;
client_max_body_size 5M;

# Frontend (React SPA)
location / {
try_files $uri $uri/ /index.html;
}

# Backend API proxy
location /api/ {
proxy_pass http://127.0.0.1:8005/api/;
proxy_http_version 1.1;
proxy_set_header Host $host;
proxy_set_header X-Real-IP $remote_addr;
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
proxy_set_header X-Forwarded-Proto $scheme;
proxy_read_timeout 120s;
}

# Uploaded files (avatars)
location /uploads/ {
proxy_pass http://127.0.0.1:8005/uploads/;
proxy_set_header Host $host;
}
}
```

```
# Enable site
sudo ln -s /etc/nginx/sites-available/hr-portal /etc/nginx/sites-enabled/

# Remove default site
sudo rm -f /etc/nginx/sites-enabled/default

# Test and restart
sudo nginx -t
sudo systemctl restart nginx
```

Step 9: Setup SSL (HTTPS) with Certbot

```
# Install Certbot
sudo apt install -y certbot python3-certbot-nginx

# Get SSL certificate (replace with your domain)
sudo certbot --nginx -d hr.sparkcurv.in

# Auto-renewal (already set up by certbot)
sudo certbot renew --dry-run
```

■ *You must point your domain DNS (A record) to EC2 public IP before running certbot*

Step 10: Verify Everything Works

Check all services:

```
# Backend
sudo systemctl status hr-backend

# Nginx
sudo systemctl status nginx

# MariaDB
sudo systemctl status mariadb

# Test API
curl -X POST http://localhost:8005/api/auth/login \
-H "Content-Type: application/json" \
-d '{"email":"ponish.jino@sparkcurv.com","password":"Aiden@1996"}'

# Should return JSON with user info
```

Open in browser:

Visit **https://hr.sparkcurv.in** (or http://your-ec2-ip)

Login with admin credentials from your .env file

Updating the Application

When you push new code to GitHub, run these on EC2:

```
cd /var/www/sparkcurv

# Pull latest code
git pull

# Rebuild frontend
cd frontend
yarn install
yarn build

# Update backend
cd ../backend
source venv/bin/activate
pip install -r requirements.txt

# Restart backend
sudo systemctl restart hr-backend
```

Troubleshooting

Backend won't start:

```
# Check logs
sudo journalctl -u hr-backend -n 50 --no-pager

# Common fixes:
sudo chown -R www-data:www-data /var/www/sparkcurv/
sudo systemctl restart hr-backend
```

Database connection error:

```
# Check MariaDB is running
sudo systemctl status mariadb

# Test connection
mysql -u root -p -h 127.0.0.1 hr_portal -e "SELECT 1;"

# Check .env file values match
```

Nginx 502 Bad Gateway:

```
# Backend is not running - restart it
sudo systemctl restart hr-backend

# Check if port matches nginx config
sudo ss -tlnp | grep 8005
```

Login returns 401 Unauthorized:

Make sure FRONTEND_URL in backend/.env starts with **https://** if using SSL

This controls cookie secure flag. Without it, cookies won't work on HTTPS.